Currents in the scrape-off layer, from scratch

Charge conservation, sheaths, cross-field currents, the three circuits, how SOLPS does it, and what the currents of JOREK's reduced MHD are

Working notes for the sheath-current paper

November 2026

Abstract

These notes build the physics of scrape-off-layer (SOL) currents from the ground up, with the single goal of being able to look at a plate-current plot from a divertor simulation and know where every ampere comes from and where it goes. Part I introduces the pieces: charge conservation, parallel Ohm's law, the sheath as a nonlinear resistor, and the currents that cross magnetic field lines. Part II assembles them into the three circuits found in the literature: Pfirsch–Schlüter, thermoelectric and plate-closing currents, plus the single-plate radial loop that dominates our own simulation. Part III shows how SOLPS (B2.5) writes and closes these currents, term by term from its manual and sources. Part IV derives, from the perpendicular momentum balance, what currents JOREK's reduced-MHD vorticity equation contains and how they map onto the SOLPS list, and gives the numbers for the AUG #33616 case. Part V reads our results with these tools. Every equation that matters is derived; every number is traceable.

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How to read these notes

Boxes marked **ELI5** say the same thing as the surrounding text without equations. Boxes marked **Key** are the statements to remember. Boxes marked **Watch out** are places where it is easy to fool yourself.

Sign conventions used throughout

- $j_n > 0$ means *ions* are collected by the wall, i.e. conventional current flows from the plasma into the wall, at *both* plates. This is the convention of JOREK's `Jn_par` $= -$`zj` $B_n/(\mu_0 F_0) = \mathbf{J}_\parallel \cdot \mathbf{n}$ with the outward normal, and of our plots. Rozhansky et al. [4] use the *tube* convention instead: positive means current flowing along the field from the inner to the outer target, i.e. "to the plate" at the outer and "away from the plate" at the inner target. A plate-to-plate (thermoelectric) current has the same sign at both plates in their convention and opposite signs in ours; a plate closing current the reverse. Translate before comparing signs.
- Along a flux tube, s runs from the inner plate ($s = 0$) to the outer plate ($s = L$), and a tube current $I(s) > 0$ flows from inner to outer. Hence $I(0) = -I_{\text{in}}$ and $I(L) = +I_{\text{out}}$ in terms of the plate currents $I_{\text{in}}, I_{\text{out}}$ (ions positive).
- Φ is the plasma potential at the sheath entrance, the wall is at V_w (grounded: $V_w = 0$).
- Temperatures in eV unless k_B is written. $\Lambda \simeq 3$ is the floating sheath drop in units of T_e/e.

Part I
The pieces

1 Charge conservation

A plasma is quasi-neutral: on scales larger than the Debye length and slower than the plasma frequency, the electron and ion charge densities are equal to a precision that no transport process can disturb. There is therefore no charge accumulation anywhere, and the current density obeys

$$\nabla \cdot \mathbf{J} = 0 \qquad \text{everywhere, at every instant.} \tag{1}$$

This is the whole content of "current conservation". It does not say that current flows along field lines; it says that whatever flows into a volume flows out of it.

ELI5: water in pipes

Think of current as water in a network of pipes that cannot store water. Whatever goes into a junction comes out of it. A *flux tube* is one pipe with two open ends, the two divertor plates, but its walls are not watertight: water can also leak through the sides into the neighbouring pipes. The statement "what goes in at one plate comes out at the other" is only true if the walls are watertight. The whole question of this document is how leaky the walls are and what makes them leak.

Integrate (1) over the volume of a flux tube bounded by two flux surfaces ψ_a, ψ_b and the two plates (Fig. 1). The divergence theorem gives

$$\boxed{I_{\text{in}} + I_{\text{out}} + I_{\text{sides}} = 0} \tag{2}$$

where $I_{\rm in}, I_{\rm out}$ are the currents into the two plates (ions positive) and $I_{\rm sides}$ is the current leaving through the two flux surfaces. The "rule" $I_{\rm in} = -I_{\rm out}$ is (2) with $I_{\rm sides} = 0$. Only currents with a component along $\nabla\psi$ contribute to $I_{\rm sides}$. The parallel current never does, since $\mathbf{B}\cdot\nabla\psi = 0$. So the rule holds exactly if and only if there is no cross-field current, and the rest of these notes is about the cross-field currents.

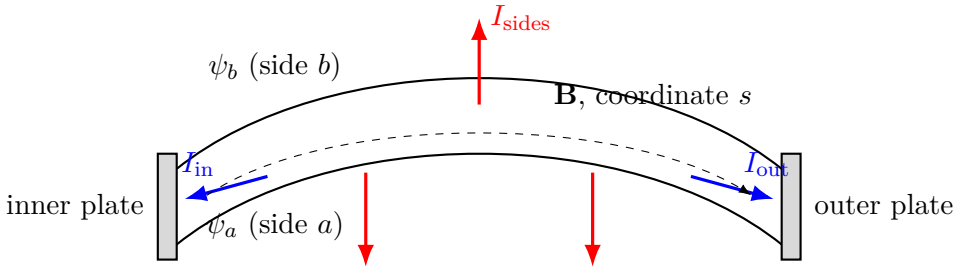


Figure 1: The control volume behind Eq. (2): a flux tube between two flux surfaces, ending on the two divertor plates. Blue: plate currents. Red: current through the sides, which only cross-field currents can carry.

2 Parallel current: Ohm's law along the field

Electrons are light and move freely along $\mathbf{B}$. Their parallel momentum balance, neglecting inertia, is the generalised Ohm's law:

$$E_\parallel \;=\; \eta_\parallel\, j_\parallel \;-\; \frac{1}{en}\,\nabla_\parallel p_e \;-\; \frac{0.71}{e}\,\nabla_\parallel T_e. \tag{3}$$

The three terms are: the resistive drop, the electron pressure gradient (electrons pushed along the field by their own pressure), and the thermal force (electron–ion friction depends on velocity, so a temperature gradient drags electrons toward the cold end with coefficient 0.71 for $Z = 1$). The parallel Spitzer resistivity is

$$\eta_\parallel \simeq 5.2\times10^{-5}\,\frac{Z\,\ln\Lambda_c}{T_e[\mathrm{eV}]^{3/2}}\;\Omega\,\mathrm{m}, \tag{4}$$

T_e	10 eV	20 eV	50 eV	100 eV
$\eta_\parallel$ [Ωm]	2.5×10^{-5}	8.7×10^{-6}	2.2×10^{-6}	7.8×10^{-7}

No current: the Boltzmann relation. Put $j_\parallel = 0$ in (3) and write $E_\parallel = -\nabla_\parallel\Phi$. For isothermal electrons,

$$\Phi(s) - \Phi(0) = \frac{T_e}{e}\,\ln\frac{n(s)}{n(0)}. \tag{5}$$

This matters enormously for reading potentials: *two ends of a tube can differ by tens of volts with zero current flowing*, if their densities and temperatures differ. A potential difference is not a drive for current; only the departure from (3) with $j_\parallel = 0$ is.

Resistance of a tube. For a current I through a tube of cross-section $A(s)$ and length L along the field,

$$R_{\rm tube} = \int_0^L \frac{\eta_\parallel(s)}{A(s)}\,\mathrm{d}s. \tag{6}$$

For a SOL tube $L \sim 50$–$100\,\mathrm{m}$, so even a good conductor is a long wire.

3 The sheath: a nonlinear resistor at each end

At a material surface the electrons, being faster, would leave first. The plasma charges positive until the resulting potential drop in a thin layer (the sheath, a few Debye lengths) repels enough electrons for the two fluxes to balance. Two standard results [1]:

$$j_{i,n} = e\,n\,c_s\,|\mathbf{b}\cdot\mathbf{n}|, \qquad c_s = \sqrt{\frac{\gamma\,(T_e + T_i)}{m_i}} \qquad \text{(Bohm: ions arrive at the sound speed)} \tag{7}$$

$$j_{e,n} = e\,n\,\frac{\bar{v}_e}{4}\,|\mathbf{b}\cdot\mathbf{n}|\,\exp\Big(-\frac{e(\Phi - V_w)}{T_e}\Big), \qquad \bar{v}_e = \sqrt{\frac{8T_e}{\pi m_e}} \quad \text{(Boltzmann-reduced electron flux)} \tag{8}$$

The factor $|\mathbf{b}\cdot\mathbf{n}|$ projects the parallel flux onto the wall normal; at grazing incidence (2–3° on divertor targets) it is 0.03–0.05.

Floating potential. With $j_{i,n} = j_{e,n}$,

$$\frac{e(\Phi - V_w)}{T_e} = \Lambda \equiv \ln\frac{\bar{v}_e}{4c_s} = \tfrac{1}{2}\ln\Big[\frac{m_i}{2\pi m_e\,\gamma(1 + T_i/T_e)}\Big] \simeq 2.6\text{–}2.8\ \text{(D)}, \tag{9}$$

and JOREK uses the round constant $\Lambda = 3$. A floating plate point therefore sits at $\Phi \simeq 3\,T_e/e$ above the wall: 30 V at 10 eV, 165 V at 55 eV.

Current–voltage characteristic. With a net current, $j_n = j_{i,n} - j_{e,n}$:

$$\boxed{j_n = j_{\rm sat}\,[1 - e^{\,x}], \qquad x = \Lambda - \frac{e(\Phi - V_w)}{T_e}, \qquad j_{\rm sat} = e\,n\,c_s\,|\mathbf{b}\cdot\mathbf{n}|} \tag{10}$$

Fig. 2 shows it. Three regimes: Φ above floating ($x < 0$) collects ions, saturating at $j_{\rm sat}$; Φ at floating carries nothing; Φ below floating ($x > 0$) collects electrons, exponentially, up to electron saturation at $x = \Lambda$ where $j_n = -(e^{\Lambda} - 1)j_{\rm sat} \simeq -19\,j_{\rm sat}$.

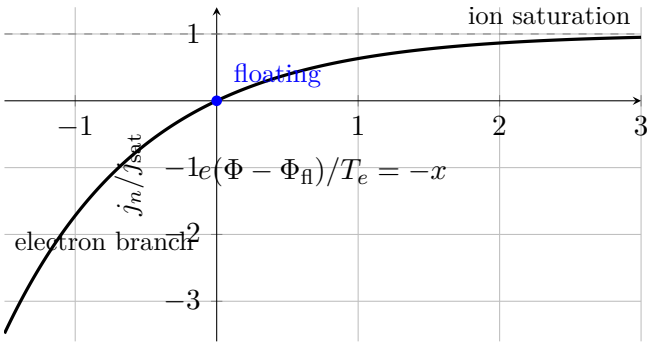


Figure 2: The sheath characteristic, Eq. (10). Ion current is positive. A few T_e/e either side of floating change the current by order $j_{\rm sat}$: the sheath is a strongly nonlinear resistor.

Small-signal sheath resistance. Linearising (10) around floating, $j_n \simeq -j_{\rm sat}\,x = j_{\rm sat}\,(e\,\delta\Phi/T_e)$, so per unit area

$$R_{\rm sh} = \frac{T_e}{e\,j_{\rm sat}} \quad [\Omega\,\mathrm{m}^2], \qquad R_{\rm sh}^{\rm (tube)} = \frac{T_e}{e\,I_{\rm sat}} \quad [\Omega], \tag{11}$$

with $I_{\rm sat}$ the ion saturation current of the tube's footprint. For a tube that can deliver 1 kA of ion saturation current at 20 eV, $R_{\rm sh} = 0.02\,\Omega$. This is the same order as the Spitzer resistance of a 70 m SOL tube, which is why the sheaths cannot be dropped from the thermoelectric circuit.

ELI5: the sheath

The plasma in front of a metal wall behaves like a diode with a built-in offset: it sits by itself at about three times its electron temperature above the wall, in volts. Push it a little higher and ions pour in, up to a limit. Pull it a little lower and electrons pour in, with no practical limit. Whoever sets the potential in front of the plate therefore sets the current through it, and the plate, being metal, cannot be at two potentials at once.

4 Currents across the magnetic field

Charged particles do not cross field lines freely, but currents across the field exist whenever there is a force on the fluid perpendicular to $\mathbf{B}$. Take the ion perpendicular momentum balance, keeping inertia, pressure, viscosity and friction on neutrals:

$$\rho \frac{\mathrm{d}\mathbf{v}_\perp}{\mathrm{d}t} = \mathbf{J} \times \mathbf{B} - \nabla p + \nabla \cdot \boldsymbol{\Pi} - \rho\, \nu_{in}\, (\mathbf{v} - \mathbf{v}_n). \tag{12}$$

Cross with $\mathbf{B}$ and divide by B^2 to solve for the perpendicular current:

$$\mathbf{J}_\perp = \underbrace{\frac{\mathbf{B} \times \nabla p}{B^2}}_{\mathbf{j}_{\mathrm{dia}}} + \underbrace{\frac{\rho}{B^2}\, \mathbf{B} \times \frac{\mathrm{d}\mathbf{v}_\perp}{\mathrm{d}t}}_{\mathbf{j}_{\mathrm{pol}}} - \underbrace{\frac{\mathbf{B} \times \nabla \cdot \boldsymbol{\Pi}}{B^2}}_{\mathbf{j}_{\mathrm{vis}}} + \underbrace{\frac{\rho \nu_{in}}{B^2}\, \mathbf{B} \times (\mathbf{v} - \mathbf{v}_n)}_{\mathbf{j}_{in}} \tag{13}$$

Every cross-field current in SOLPS and in JOREK is one of these four, or an ad hoc addition ("anomalous conductivity"). They are driven by *forces*, not by the electric field directly: the plasma is not an ohmic conductor across $\mathbf{B}$. Fig. 3 shows the single-particle origin of the first and most important one.

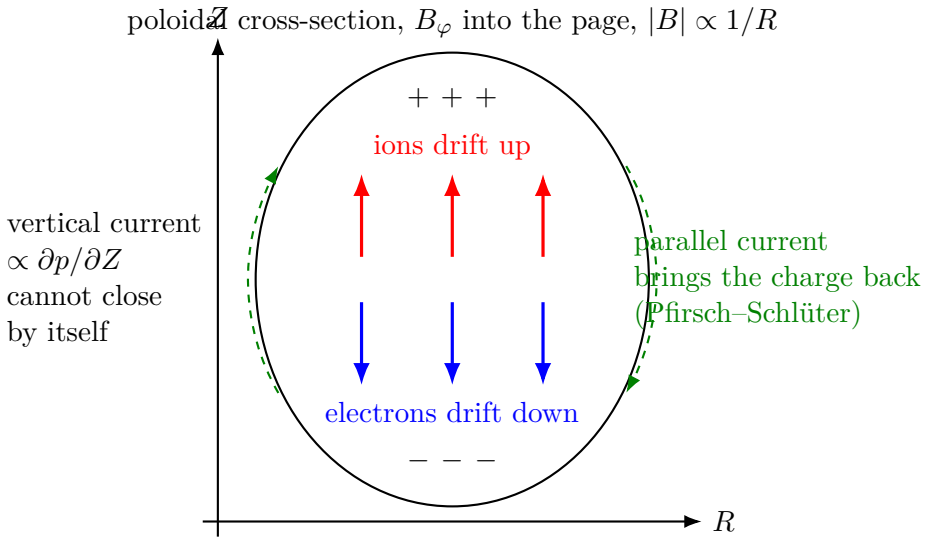


Figure 3: The grad-B and curvature drifts separate charges vertically. The resulting vertical current is the divergence-carrying part of the diamagnetic current. On closed surfaces it is closed by parallel current; on open field lines that parallel current ends on the plates.

4.1 The diamagnetic current and its divergence

$\mathbf{j}_{\mathrm{dia}} = \mathbf{B} \times \nabla p / B^2$ is perpendicular to both $\mathbf{B}$ and ∇p. On a closed surface with $p = p(\psi)$ it lies inside the flux surface and carries nothing across it. Its divergence, however, is not zero in a torus.

Using $\nabla\cdot(\mathbf{a}\times\mathbf{c}) = \mathbf{c}\cdot\nabla\times\mathbf{a} - \mathbf{a}\cdot\nabla\times\mathbf{c}$ with $\nabla\times\nabla p = 0$,

$$\nabla\cdot\mathbf{j}_{\text{dia}} = \nabla p\cdot\nabla\times\Big(\frac{\mathbf{B}}{B^2}\Big). \tag{14}$$

For a tokamak $\mathbf{B} \simeq (F_0/R)\,\mathbf{e}_\varphi$ to leading order, so $\mathbf{B}/B^2 \simeq (R/F_0)\,\mathbf{e}_\varphi$, and in cylindrical coordinates $\nabla\times(A_\varphi\mathbf{e}_\varphi) = -\partial_Z A_\varphi\,\mathbf{e}_R + \frac{1}{R}\partial_R(RA_\varphi)\,\mathbf{e}_Z$ gives $\nabla\times(R\mathbf{e}_\varphi/F_0) = (2/F_0)\,\mathbf{e}_Z$ up to the orientation of φ. Hence

$$\boxed{\nabla\cdot\mathbf{j}_{\text{dia}} = \pm\frac{2}{F_0}\frac{\partial p}{\partial Z} = \pm\frac{2}{BR}\frac{\partial p}{\partial Z}} \tag{15}$$

with the upper sign in a right-handed (R,φ,Z) basis and the lower sign in JOREK's right-handed (R,Z,φ) basis, where $\mathbf{e}_\varphi\times\nabla p = (-\partial_Z p,\,+\partial_R p)$ and $\mathbf{j}_{\text{dia}} = (R/F_0)(-\partial_Z p,\,\partial_R p)$, which is also what the postproc's `JR`, `JZ` contain. This is the "vertical ∇B current" of the SOL literature [3, 6]: wherever the pressure varies vertically, charge is being deposited at the rate (15) and something else must carry it away. On a closed surface the vertical variation of $p(\psi)$ along the surface is what drives the Pfirsch–Schlüter current (Sec. 5). On open field lines the pressure falls along the divertor legs, which are nearly vertical, so the legs are strong sources of (15).

Watch out: "the diamagnetic current crosses flux surfaces" is true on open lines

With $p = p(\psi)$ the diamagnetic current is tangent to flux surfaces. In the SOL, p also varies *along* the field line, towards the plates. The poloidal component of that gradient, crossed with the toroidal field, gives a radial diamagnetic current, $j_{\text{dia},r} \simeq -(B_\varphi/B^2)\,\partial p/\partial s_{\text{pol}}$. In a flux-tube budget it may enter through one side and leave through the other; its *net* deposit is (15) integrated over the tube.

4.2 Polarisation current

The term $\rho\,\mathrm{d}\mathbf{v}_\perp/\mathrm{d}t$ in (12) with $\mathbf{v}_\perp \simeq \mathbf{E}\times\mathbf{B}/B^2$ gives

$$\mathbf{j}_{\text{pol}} = \frac{\rho}{B^2}\frac{\mathrm{d}\mathbf{E}_\perp}{\mathrm{d}t} = \frac{\rho}{B^2}\Big(\partial_t + \mathbf{v}_E\cdot\nabla\Big)\mathbf{E}_\perp, \tag{16}$$

the ion inertia responding to a changing perpendicular electric field. It has a time-derivative part, important in transients (ELMs, a drifting equilibrium) and an advective part, the Reynolds stress of the $E\times B$ flow, which is present in steady state wherever the flow is sheared. The gyroviscous cancellation adds the diamagnetic velocity to the advecting flow; the structure stays the same.

4.3 Viscous current

A viscous stress $\mathbf{\Pi} = -\mu\nabla\mathbf{v}$ exerts a force $\mu\nabla^2\mathbf{v}$; its cross-field current is $\mathbf{j}_{\text{vis}} = -\mathbf{B}\times\mu\nabla^2\mathbf{v}_\perp/B^2$. For the $E\times B$ flow, $\mathbf{v}_E = \mathbf{B}\times\nabla\Phi/B^2$, so

$$\nabla\cdot\mathbf{j}_{\text{vis}} \;\simeq\; \frac{\mu}{B^2}\,\nabla_\perp^2(\nabla_\perp^2\Phi)\cdot\tfrac{1}{B}\cdot B \;\propto\; \mu\,\nabla_\perp^2\,\omega, \qquad \omega = \nabla_\perp^2\Phi/B, \tag{17}$$

a fourth derivative of the potential: viscosity moves charge wherever the vorticity of the $E\times B$ flow is curved. Shear layers, such as the E_r well at the separatrix, are where this current lives. Its size is set by μ, which in both SOLPS and JOREK is an anomalous (input) coefficient.

4.4 Friction and anomalous currents

Ion–neutral friction gives $\mathbf{j}_{in} \propto \rho\nu_{in}\mathbf{B}\times\mathbf{v}/B^2$, important only in cold, neutral-rich divertors. SOLPS additionally allows an *anomalous perpendicular conductivity*, $\mathbf{j}_{AN} = -\sigma_{AN}\nabla_\perp\Phi$, an ohmic term with no physical derivation, kept as a knob and usually set small [3].

Key: what carries current across the field

Diamagnetic (pressure), polarisation (inertia and Reynolds stress), viscous (shear), ion–neutral friction, and in SOLPS an optional anomalous ohmic term. None is proportional to the electric field alone. The diamagnetic current is usually the largest, but on a flux-tube budget it often passes straight through; the *net* cross-field current into a tube is a competition between its divergence (15) and the inertial and viscous terms.

Part II
The circuits

5 Pfirsch–Schlüter current on closed surfaces

On a closed flux surface with $p = p(r)$, Eq. (1) with (15) requires a parallel current whose divergence cancels the vertical deposition:

$$\mathbf{B}\cdot\nabla\Big(\frac{j_\parallel}{B}\Big) = -\nabla\cdot\mathbf{j}_{\rm dia} = -\frac{2}{BR}\,\frac{\partial p}{\partial Z}. \tag{18}$$

In a large-aspect-ratio circular geometry, $Z = r\sin\theta$, $\partial p/\partial Z = p'(r)\sin\theta$, $\mathbf{B}\cdot\nabla = (B_\theta/r)\,\partial_\theta$, $B \simeq B_0$, $R \simeq R_0$:

$$\frac{B_\theta}{r}\,\partial_\theta\Big(\frac{j_\parallel}{B_0}\Big) = -\frac{2p'\sin\theta}{B_0 R_0} \quad\Longrightarrow\quad \boxed{j_\parallel^{PS} = \frac{2\,q}{B_0}\,p'(r)\,\cos\theta}, \qquad q = \frac{r B_0}{R_0 B_\theta}. \tag{19}$$

It is anti-parallel on the outboard side ($p' < 0$), parallel on the inboard side, averages to zero over the surface, and is q times larger than the diamagnetic current it closes. Fig. 3 is the picture. Nothing crosses the flux surface: the PS current only redistributes charge *within* it. That is why, in a flux-surface average, the parallel current drops out of charge conservation identically.

6 Open field lines: currents that end on the plates

In the SOL, integrating (18) along the field from the inner plate gives

$$\frac{j_\parallel(s)}{B} = \frac{j_\parallel(0)}{B} - \int_0^s \frac{2}{B^2 R}\,\frac{\partial p}{\partial Z}\,\mathrm{d}s'. \tag{20}$$

Now the current does not have to average to zero: whatever the integral accumulates arrives at the plates. Two things follow.

The PS current reaches the targets. Above the X-point the SOL tubes still see p varying along the surface and the PS pattern of (19) forms, but it is now absorbed by the sheaths instead of closing on itself. Schaffer, Chankin et al. measured exactly such current layers at the JET target and found their magnitude and sign fit the SOL PS current rather than a thermoelectric current [8].

The legs are sources. Below the X-point the pressure drops along the near-vertical legs, so $\partial p/\partial Z$ is large and (15) deposits charge in the legs. Zhao et al. write this source as $\partial j_\parallel/\partial s \simeq (1/BR)\,\partial p_{\rm tot}/\partial Z$ and note that the resulting parallel current "flows away from the location of the strongest source toward both divertor targets" [6]. Both targets: same sign at the two plates. Rozhansky et al. call these *plate closing currents of type 1* [4].

7 The thermoelectric circuit (Staebler–Hinton)

Now combine Ohm's law (3) along a tube with a sheath at each end [2]. Let the tube current be I, positive from inner to outer, so the inner plate collects $-I$ and the outer $+I$ (ions positive). Linearise both sheaths around floating with (11):

$$\Phi_{\rm in} = \frac{T_e^{\rm in}}{e}\,\Lambda - R_{\rm sh}^{\rm in}\,I, \qquad \Phi_{\rm out} = \frac{T_e^{\rm out}}{e}\,\Lambda + R_{\rm sh}^{\rm out}\,I. \tag{21}$$

Integrate (3) from inner to outer with $E_\parallel = -\partial_s\Phi$:

$$\Phi_{\rm out} - \Phi_{\rm in} = -R_{\rm tube}\,I + \int_0^L \frac{\partial_s p_e}{en}\,{\rm d}s + \frac{0.71}{e}\,(T_e^{\rm out} - T_e^{\rm in}). \tag{22}$$

Eliminate the potentials:

$$\boxed{I = -\,\frac{\dfrac{(\Lambda - 0.71)}{e}\,(T_e^{\rm out} - T_e^{\rm in}) \;-\; \displaystyle\int_0^L \frac{\partial_s p_e}{en}\,{\rm d}s}{R_{\rm sh}^{\rm in} + R_{\rm sh}^{\rm out} + R_{\rm tube}}} \tag{23}$$

Read it as a circuit (Fig. 4): the open-circuit voltage is the difference of the two floating drops, $\Lambda\Delta T_e/e$, reduced by the thermal force and by the electron-pressure (Boltzmann) term, driving current through three resistors in series. With $T_e^{\rm out} > T_e^{\rm in}$, $I < 0$: in the plasma the current flows from the hot plate to the cold one, the hot plate collects electrons and the cold plate collects ions. This is the textbook SOL current, measured on many machines with shunts in the target tiles [9].

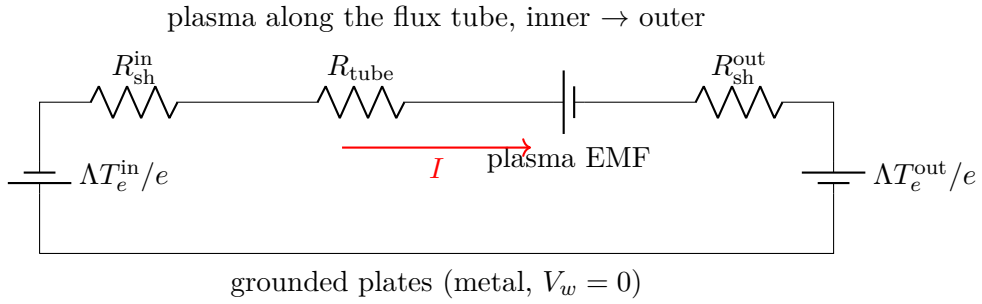


Figure 4: The thermoelectric circuit of Eq. (23). Two sheath "batteries" with their internal resistances, the tube resistance $R_{\rm tube} = \int \eta_\parallel\,{\rm d}s/A$, and the Boltzmann/thermal-force EMF of the plasma in between, $\int \partial_s p_e/(en) + 0.71\Delta T_e/e$. If the plasma EMF matches the sheath difference, no current flows even though the two plates sit at very different potentials.

Watch out: a potential difference between the plates is not a current

If both plates float and the Boltzmann term balances $\Lambda\Delta T_e/e$, then $\Phi_{\rm out} - \Phi_{\rm in} = \Lambda\Delta T_e/e$ exactly, tens of volts, with $I = 0$. In our AUG run $\Phi_{\rm out} - \Phi_{\rm in}$ equals $\Lambda\Delta T_e/e$ within a few volts on every tube: the circuit is open. The 20–40 V between the plates are held by the electrons, not dropped across a resistor.

Numbers for a SOL tube of AUG. $\Delta T_e \approx 7$ eV, $L_\parallel \approx 70$ m, footprint delivering $I_{\rm sat} \approx 1$ kA at 15–20 eV. Then $R_{\rm sh} \approx 0.02\,\Omega$ per sheath, $R_{\rm tube}^{\rm Spitzer} \approx 0.02\,\Omega$ for a cross-section of $\sim 0.05\,{\rm m}^2$, open-circuit voltage ~ 16 V, so $I_{\rm Spitzer} \sim 0.3$ kA. With a resistivity 20 times Spitzer, $R_{\rm tube} \approx 0.4\,\Omega$ and $I \approx 0.04$ kA. The thermoelectric current is proportional to $1/\eta_\parallel$ once the tube resistance dominates.

8 The single-plate radial loop

The circuit of Sec. 7 uses two plates. There is a second circuit that needs only one, and it is the one our sheath-current simulations are dominated by. Along one plate the temperature varies strongly: 55 eV at the strike point, 12 eV a few centimetres into the SOL. Each flux tube ending there would like to float at its own $\Lambda T_e/e$: 165 V at the strike point, 36 V on the wing. The plasma cannot sustain a 130 V jump across millimetres of flux surfaces: the resulting radial electric field is of order 10^5 V/m, with a sheared $E \times B$ flow whose polarisation and viscous currents (16), (17) flow down the potential hill. So $\Phi(\psi)$ settles at a compromise, and the sheath law (10) converts the compromise into current (Fig. 5):

- at the strike point $\Phi < \Lambda T_e/e$: electrons flow into the plate, $j_n < 0$;
- on the wings $\Phi > \Lambda T_e/e$: ions flow into the plate, $j_n > 0$.

Conventional current enters the plasma at the strike point, flows radially outward across the flux surfaces in the SOL, re-enters the plate on the wings, and returns through the metal. Both plates have a hot spot and wings, so both show the same sign pattern, and the two plate profiles against ψ have the same sign everywhere, which is the signature that distinguishes this loop from the thermoelectric one. In the taxonomy of [4] this is a plate closing current of type 2: a radial current in the SOL above the X-point, closed through the plates. What sets its magnitude is the cross-field conductance of the shear layer, i.e. the viscosity and inertia terms, not the parallel resistivity.

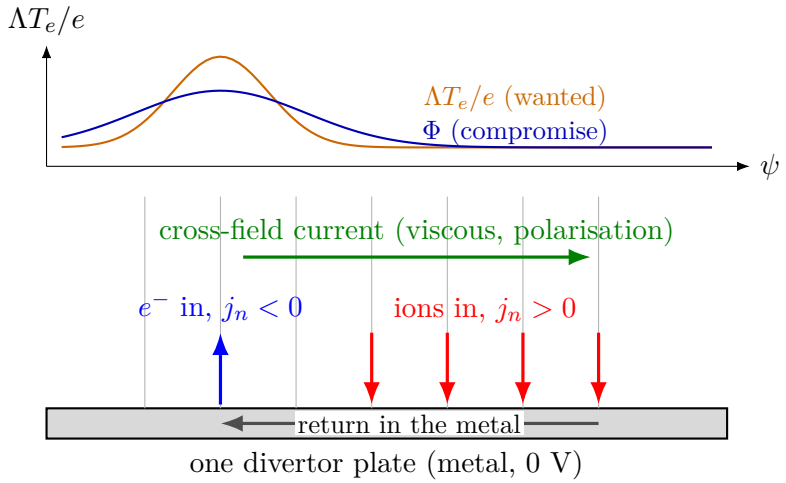


Figure 5: The single-plate radial loop. The plate is one conductor; the plasma in front of it wants a different floating potential on every flux surface. Cross-field currents smooth $\Phi(\psi)$, the sheath law turns the mismatch into electron collection at the hot spot and ion collection on the wings, and the metal closes the loop.

9 Putting the families together

Rozhansky et al. [4], from SOLPS-ITER simulations of ASDEX Upgrade and Globus-M, state that the measured plate currents are the sum of three families (Table 1). The single-plate loop of Sec. 8 is their type 2 plate closing current viewed from the plate. Fig. 6 sketches where each flows.

What the per-tube sum measures. Returning to (2): the thermoelectric and PS parts have opposite signs at the two ends and cancel in $I_{\text{in}} + I_{\text{out}}$; the plate closing parts have the same sign and add. So

$$\begin{aligned} \tfrac{1}{2}\,(I_{\text{out}} - I_{\text{in}}) &= \text{current through the tube (thermoelectric + PS reaching the plates)}, \\ I_{\text{in}} + I_{\text{out}} &= \text{plate closing current} = -I_{\text{sides}}. \end{aligned} \tag{24}$$

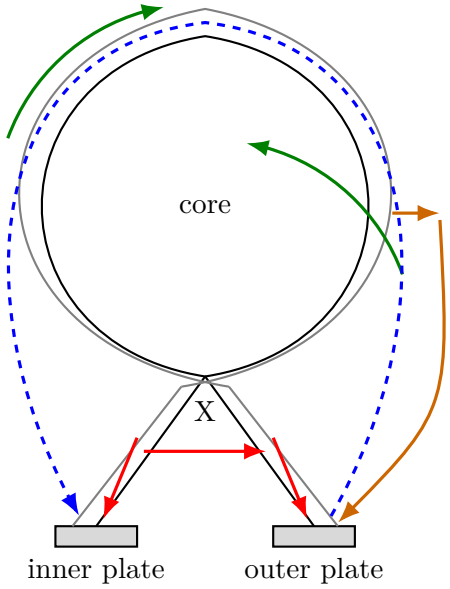
family	drive	path	sign at the two plates
Pfirsch–Schlüter	$\partial p/\partial Z$ on a surface above the X-point	along the surface; on open lines ends on the plates	opposite (it is a parallel current pattern)
thermoelectric	$\Lambda\Delta T_e$ between the plates minus the plasma EMF	along the tube, plate to plate	opposite: hot plate electrons, cold plate ions
plate closing, type 1	vertical ∇B current of the legs, Eq. (15)	leg $\rightarrow$ both plates	same
plate closing, type 2	radial currents in the SOL above the X-point (viscous, inertial, anomalous)	SOL $\rightarrow$ both plates	same

Table 1: The current families of the SOL after [4, 5]. In the SOL the type-1 PCC adds to the thermoelectric current, in the private region it subtracts from it.

The colleague's rule says the second is zero. In a code with cross-field currents it is a measurement of them.

10 What experiments see

- **Thermoelectric currents** are routinely measured with shunts in the divertor tiles. On ASDEX Upgrade they flow from the hotter outer to the colder inner target in attached L- and H-mode and reverse when the outer target detaches first; they are large enough to matter for the power balance [9].
- **Pfirsch–Schlüter layers** were identified at the JET target by Schaffer, Chankin et al.: narrow current layers whose sign and size fit the SOL PS current, passing through the recycling zone and the sheath [8].
- **ELM currents**: on TCV, Pitts et al. measured target currents during ELMs comparable to the ion saturation current, with current, floating potential and T_e arriving in that order [10].
- **Detachment**: SOLPS-ITER work by Senichenkov et al. finds that when both divertors detach the thermal current collapses and the net plate current drops to the size of the radial current leaving through the outer SOL and private-flux boundaries of the simulation domain [7].
- **Closing versus thermoelectric**: in UEDGE simulations of DIII-D detachment, Zhao et al. estimate the thermoelectric current an order of magnitude below the parallel closing current of the ∇B source [6].



PS: along the surface,
closes on itself above the X-point

PCC 2: radial current in the SOL,
closed through the plates

thermoelectric:
hot plate → cold plate
along the tube (dashed)

PCC 1: ∇B current
across the legs → both plates

Figure 6: Where the families flow, schematically. Green: Pfirsch–Schlüter pattern. Blue: thermoelectric current from the hotter to the colder plate. Red: plate closing current of type 1, the vertical ∇B current of the legs returning through both plates. Orange: type 2, a radial SOL current above the X-point returning through the plates.

Part III
How SOLPS does it

11 The B2.5 current equation

SOLPS (B2.5) solves the current continuity equation (1) for the electrostatic potential on its (x, y) grid, x along and y across the flux surfaces (manual eq. B.13 / C.191):

$$\frac{1}{\sqrt{g}}\frac{\partial}{\partial x}\Big(\frac{\sqrt{g}}{h_x}\,\tilde{j}_x\Big) + \frac{1}{\sqrt{g}}\frac{\partial}{\partial y}\Big(\frac{\sqrt{g}}{h_y}\,\tilde{j}_y\Big) = 0, \qquad \tilde{j}_x = b_z\,\tilde{j}_\perp + b_x\,j_\parallel. \tag{25}$$

The tilde means "effective": divergence-free parts of the diamagnetic current are removed analytically because they cannot be represented stably on the grid, keeping only the part with the same divergence [3]. The radial current (manual C.211–C.229) is the sum

$$j_y = j_y^{(AN)} + \tilde{j}_y^{(dia)} + j_y^{(in)} + \tilde{j}_y^{(vis\parallel)} + j_y^{(vis\perp)} + \tilde{j}_y^{(visq)} + \tilde{j}_y^{(s)} + j_y^{(ST)}, \tag{26}$$

with, for a single main ion species,

$$j_y^{(AN)} = -\sigma^{(AN)}\,\frac{\partial\Phi}{h_y\partial y}, \qquad \sigma^{(AN)} = (c_{\sigma,0} + c_{\sigma,1}\tfrac{10^{20}}{n_e} + \dots)\,e\,n_e,\ c_{\sigma,0}\sim 10^{-6}\text{–}10^{-3} \tag{27}$$

$$\tilde{j}_y^{(dia)} = -B_z\,n\,(T_e+T_i)\,\frac{\partial}{h_x\partial x}\Big(\frac{1}{B^2}\Big), \qquad \text{(vertical } \nabla B \text{ current, cf. (15))} \tag{28}$$

$$j_y^{(in)} = -\frac{B_z}{2}\,\frac{\partial}{h_x\partial x}\Big(\frac{1}{B^2}\Big)\,m\,n\,V_\parallel^2, \qquad \text{(inertia of the parallel flow)} \tag{29}$$

$$j_y^{(vis\perp)} = -\frac{1}{B\sqrt{g}}\,\frac{\partial}{\partial y}\Big(\frac{\sqrt{g}}{h_y}\,\frac{\eta^{(AN)}}{4}\,\frac{\partial\tilde{V}_\perp}{h_y\partial y}\Big), \qquad V_\perp = -\frac{1}{Bh_y}\frac{\partial\Phi}{\partial y} - \frac{1}{enBh_y}\frac{\partial nT_i}{\partial y} \tag{30}$$

$$\tilde{j}_y^{(s)} = -\sigma_{in}\Big(\frac{\partial\Phi}{h_y\partial y} + \frac{1}{en}\frac{\partial nT_i}{h_y\partial y} + BV_{n\perp}\Big), \qquad \sigma_{in} = \frac{\tilde{\mu}\,m_p\,n\,n_n\langle V\sigma_{cx}\rangle}{B^2} \tag{31}$$

plus the parallel-viscosity and heat-viscosity currents (small corrections, needed on closed surfaces to make the surface-averaged radial current vanish) and an optional stochastic-layer conductivity. Term by term this is Eq. (13): (28) is the diamagnetic current written through $\partial(1/B^2)$, which with $B \propto 1/R$ is the same $\partial p/\partial Z$ source as (15); (29) is the inertial term; (30) the perpendicular viscous current acting on the $E\times B$ plus diamagnetic flow, with an anomalous viscosity $\eta^{(AN)} = mnD^{(AN)}$; (31) the ion–neutral friction; (27) the anomalous conductivity. The parallel current is algebraic, from Ohm's law (3) with Spitzer (or Zhdanov) conductivity; there is no induction equation.

12 Boundary conditions for the potential

The potential equation needs a current condition on every boundary (manual C.7.3 and the routine `b2stbc_phys.F`).

Targets (west/east). `BCPOT = 3` or `11` impose the sheath characteristic as the current flux through the target face,

$$\frac{\sqrt{g}}{h_x}\, j_x\Big|_W = \frac{\sqrt{g}}{h_x}\, e\Big[\sum_a z_a \Gamma_{ax} + (1-\gamma_e)\,|b_x|\, n_e\, \frac{1}{\sqrt{2\pi}}\sqrt{\frac{T_e}{m_e}}\,\exp\Big(-\frac{e(\Phi - p_{\Phi,2})}{T_e}\Big)\Big]_W, \tag{32}$$

i.e. ion flux minus Boltzmann-reduced electron flux: exactly (10) in flux form, linearised in Φ in the guard cell. `BCPOT = 11` is the version consistent with the drift-compatible Bohm conditions (`BCCON = 14`, `BCMOM = 13`, `BCENE = 15`).

Radial boundaries (north = outer SOL edge, south = core and private-flux floor). There is no sheath. The options are: prescribed potential (1), prescribed gradient (2), prescribed current density (5) or total current (6, 8, 10 with feedback), decay length (9), and the options 12 and 13 that add the diamagnetic, inertial and other perpendicular currents of the adjacent cell to the boundary flux, so that the physical cross-field current is allowed to leave; which of these contributions are included is governed by `b2stbc_istyle_cur_contr_on_S_and_N` (default 2). At the core boundary the usual choice is zero net current.

Key: why SOLPS budgets always close

Every current in (26) and every boundary condition is a face flux of a finite-volume cell, and the plate current is itself the face flux (32). The sum of all boundary currents is therefore zero to round-off, *and each boundary's share is a printed number*. Per flux tube, however, $I_{\text{in}} + I_{\text{out}} \neq 0$ whenever (26) is non-zero on the tube's sides, which is the normal case near the separatrix.

13 What SOLPS results look like

Rozhansky's 2001 paper established that the potential near the separatrix is set by the effective perpendicular conductivity, mostly the diamagnetic current balanced by parallel viscosity, inertia and ion-neutral friction, while far out in the SOL it is set by the electron parallel force balance, i.e. Boltzmann and the sheaths, and is a weak function of the cross-field conductivity [3]. Target-current comparisons with ASDEX Upgrade probes only fit when all drifts, and with them all closing currents, are switched on [11]. In single-null attached AUG plasmas the thermoelectric current usually dominates the plate totals, with the plate closing currents as the second component; in connected double-null the thermoelectric drive is weak and the plate closing currents take over [4]. In double detachment the thermal current collapses and the plate totals shrink to the radial boundary currents [7].

Part IV
The currents of JOREK's reduced MHD

14 Fields, variables and units

Model600 (and the other reduced-MHD models) represents the magnetic field and the velocity as

$$\mathbf{B} = F_0\nabla\varphi + \nabla\psi\times\nabla\varphi, \qquad \mathbf{v} = \mathbf{v}_E + V_\parallel\mathbf{B} + \mathbf{v}_{\rm dia}, \qquad \mathbf{v}_E = -R\,\nabla u\times\mathbf{e}_\varphi, \qquad \Phi = F_0\,u, \tag{33}$$

with $F_0 = RB_\varphi$ constant, ψ the poloidal flux per radian, and u the stream function of the $E\times B$ flow. The toroidal current is carried by the variable j:

$$j \equiv \Delta^*\psi = R\,\nabla\cdot\Big(\frac{\nabla\psi}{R}\Big) = -\mu_0 R\,J_\varphi, \qquad J_\parallel = J_\varphi\,\frac{R|B|}{F_0} \;\Rightarrow\; \frac{J_\parallel}{|B|} = -\frac{j}{\mu_0 F_0}. \tag{34}$$

Normalisation: lengths in metres, $\mathbf{B}$ in tesla, $\rho_0 = m_i n_0$ with $n_0 = 10^{20}\,\mathrm{m}^{-3}\times$ `central_density`; time in units of $t_0 = \sqrt{\mu_0\rho_0}$, velocities in $1/t_0$, pressure $p_{SI} = p/\mu_0$, current $\mathbf{J}_{SI} = \mathbf{J}/\mu_0$, potential $\Phi_{SI} = \Phi/t_0$. For AUG #33616 ($n_0 = 0.765\times10^{20}$, deuterium): $t_0 = 5.67\times10^{-7}$ s, one normalised unit of Φ is 1.76×10^6 V.

Two facts shape everything below. First, $J_\parallel$ is *not algebraic*: ψ obeys an induction equation, so the parallel current is a dynamical variable with its own time scale. Second, the perpendicular currents are *never stored*: there is no variable for $\mathbf{J}_\perp$. They exist only through their divergence, in the equation that determines u.

15 Ohm's law in JOREK: the induction equation

The ψ row of model600, in the weak form the code assembles [14], is

$$\begin{aligned}\frac{1}{R}\,\partial_t\psi = {}& \frac{\eta(T_e)}{R}\,(j - j_s - j_b) + \Big([\psi,\,u] - \frac{F_0}{R}\partial_\varphi u\Big) - \nabla\cdot(\eta_{\rm num}\nabla j)\\ & - \frac{2\tau_{\rm IC}F_0^2}{\rho_c B^2R^2}\Big([\psi,\,p_e] - \frac{F_0}{R}\partial_\varphi p_e\Big) - \frac{2\tau_{\rm IC}\,c\,F_0^2}{B^2R^2}\Big([\psi,\,T_e] - \frac{F_0}{R}\partial_\varphi T_e\Big),\end{aligned} \tag{35}$$

with the bracket $[a,\,b] = \partial_R a\,\partial_Z b - \partial_Z a\,\partial_R b$, the identity $[\psi,\,f] - (F_0/R)\partial_\varphi f = -R\,\mathbf{B}\cdot\nabla f$, and $2\tau_{\rm IC} = 1/(eF_0\sqrt{\mu_0\rho_0})$ the normalised $1/(en)$ coefficient. Reading the terms with $E_\parallel$ in mind: resistive drop (η, with the `keep_current_prof` source j_s and bootstrap j_b subtracted), the $E\times B$ advection of flux (the $-\nabla\Phi$ part of $E_\parallel$), hyper-resistivity (numerical), electron pressure gradient and thermal force with $c = 0.71$ (`thermoelectric_ohm`). This is (3) projected on the toroidal direction and integrated in time. The resistivity is

$$\eta(T) = \eta_0\Big(\frac{T}{T_0}\Big)^{-3/2} \quad (\texttt{eta_T_dependent}), \qquad \eta_{SI} = \mu_0\,\frac{\eta_0}{t_0}\Big(\frac{T}{T_0}\Big)^{-3/2}, \tag{36}$$

with T_0 the axis temperature of the initial profile.

16 Deriving the currents of the vorticity equation

The u row of model600 (`rhs_ij(var_u)`) is, in strong form [13],

$$\begin{aligned}R\,\nabla\cdot(R^2\rho\,\nabla_\perp\partial_t u) = {}& \underbrace{\tfrac{1}{2}\Big[R^2|\nabla_\perp u|^2,\,R^2\rho\Big] + \Big[R^4\rho\,\omega,\,u\Big]}_{\text{(a) inertia, Reynolds stress}} + \underbrace{[\psi,\,j] - \frac{F_0}{R}\,\partial_\varphi j}_{\text{(b) parallel current}} - \underbrace{\Big[R^2,\,p\Big]}_{\text{(c) diamagnetic}}\\ & + \underbrace{R\,\mu\,\nabla^2_{\rm pol}\omega + \mathcal{V}_{\rm num}}_{\text{(d) viscosity, hyperviscosity}} + \underbrace{\mathcal{G}[\tau_{\rm IC}, p_i, u]}_{\text{(e) gyroviscous}} + \mathcal{N}_{\rm neo} + \mathcal{S}_u, \qquad \omega = \nabla^2_{\rm pol}u.\end{aligned} \tag{37}$$

We now show that this is $F_0 R\,\nabla\cdot\mathbf{J} = 0$ with $\mathbf{J}$ the currents of Sec. 4, evaluated with the reduced-MHD field $\mathbf{B} \simeq (F_0/R)\mathbf{e}_\varphi$, so that $\mathbf{B}/B^2 \simeq (R/F_0)\mathbf{e}_\varphi$. JOREK's basis (R, Z, φ) is right-handed: $\mathbf{e}_R\times\mathbf{e}_Z = \mathbf{e}_\varphi$, $\mathbf{e}_\varphi\times\mathbf{e}_R = \mathbf{e}_Z$, $\mathbf{e}_\varphi\times\mathbf{e}_Z = -\mathbf{e}_R$, hence for a poloidal gradient $\mathbf{e}_\varphi\times\nabla f = (-\partial_Z f,\, +\partial_R f)$ in (R, Z) components. (With this orientation $B_R = \partial_Z\psi/R$, $B_Z = -\partial_R\psi/R$ and $[\psi,\, f] = -R\,\mathbf{B}_{\rm pol}\cdot\nabla f$, as in the code.)

(c) The diamagnetic current. $\mathbf{j}_{\rm dia} = \mathbf{B}\times\nabla p/B^2 = (R/F_0)\,\mathbf{e}_\varphi\times\nabla p = (R/F_0)\,(-\partial_Z p,\, \partial_R p)$. Its divergence,

$$\nabla\cdot\mathbf{j}_{\rm dia} = -\frac{1}{R}\partial_R\Big(\frac{R^2}{F_0}\partial_Z p\Big) + \partial_Z\Big(\frac{R}{F_0}\partial_R p\Big) = -\frac{2}{F_0}\,\partial_Z p \quad\Longrightarrow\quad F_0 R\,\nabla\cdot\mathbf{j}_{\rm dia} = -2R\,\partial_Z p = -\Big[R^2,\, p\Big]. \tag{38}$$

This is term (c) exactly: the code assembles it as $+R^2\,[v,\, p]$ against the test function v (`factor(var_u,4)`), whose strong form after integration by parts is $-\big[R^2,\, p\big]$. It is (15): the curvature term of the reduced-MHD vorticity equation is the vertical ∇B current source of the SOL literature. JOREK contains the drive of the Pfirsch–Schlüter current and of the type-1 plate closing current.

(b) The parallel current. $\nabla\cdot(J_\parallel\mathbf{b}) = \mathbf{B}\cdot\nabla(J_\parallel/B)$ and, from (34) in normalised units ($\mu_0 = 1$), $J_\parallel/B = -j/F_0$. With $\mathbf{B}\cdot\nabla f = -\frac{1}{R}(\,[\psi,\, f] - \frac{F_0}{R}\partial_\varphi f)$,

$$F_0 R\,\nabla\cdot(J_\parallel\mathbf{b}) = F_0 R\,\Big(-\frac{1}{F_0}\Big)\Big(-\frac{1}{R}\Big)\Big(\,[\psi,\, j] - \frac{F_0}{R}\partial_\varphi j\Big) = [\psi,\, j] - \frac{F_0}{R}\,\partial_\varphi j, \tag{39}$$

term (b). In an axisymmetric state the ∂_φ part vanishes and $[\psi,\, j]$ is $-R\,\mathbf{B}_{\rm pol}\cdot\nabla j$: the variation of the parallel current along the poloidal projection of the field line.

(lhs) and (a) The polarisation current. With $\mathbf{v}_E = -R\nabla u\times\mathbf{e}_\varphi$ and $\mathbf{B}/B^2 \simeq (R/F_0)\mathbf{e}_\varphi$, $\mathbf{B}\times\rho\,\partial_t\mathbf{v}_E/B^2 = \rho\,(R/F_0)\,\mathbf{e}_\varphi\times(\,-R\,\partial_t\nabla u\times\mathbf{e}_\varphi) = -(\rho R^2/F_0)\,\nabla_\perp\partial_t u$, using $\mathbf{e}_\varphi\times(\mathbf{a}\times\mathbf{e}_\varphi) = \mathbf{a}$ for poloidal $\mathbf{a}$. So

$$F_0 R\,\nabla\cdot\mathbf{j}_{\rm pol}^{(\partial_t)} = -R\,\nabla\cdot(\rho R^2\nabla_\perp\partial_t u), \tag{40}$$

which is minus the left-hand side of (37): moving it across gives the equation as written. The advective part $\rho\,(\mathbf{v}_E\cdot\nabla)\mathbf{v}_E$ produces, after the same manipulation and an integration by parts, the two brackets of term (a): the Reynolds stress of the $E\times B$ flow, $\big[R^4\rho\omega,\, u\big]$, and the kinetic-energy bracket. The gyroviscous block (e) corrects the advecting velocity from $\mathbf{v}_E$ to $\mathbf{v}_E + \mathbf{v}_{\rm dia}$ (the gyroviscous cancellation), as in B2.5 where inertia and gyroviscosity are combined for the same reason.

(d) The viscous current. Physically $\mathbf{j}_{\rm vis} = -\mathbf{B}\times\mu\nabla^2\mathbf{v}_E/B^2 \simeq (\mu R^2/F_0)\,\nabla_\perp\omega$ with $\omega = \nabla^2_{\rm pol}u$. The code has two forms (`visco_old_setup`). The old one, used in the AUG case, assembles $-\mu R\,\nabla v\cdot\nabla\omega$, i.e. $F_0 R\,\nabla\cdot\mathbf{j}_{\rm vis} = \nabla\cdot(\mu R\nabla\omega)$ and $\mathbf{j}_{\rm vis} = (\mu/F_0)\nabla_\perp\omega$: the R^2 of the physical form is absorbed into the definition of μ. The new one assembles $-\mu R^3\nabla v\cdot\nabla\omega - 2\mu R^2\omega\,\partial_R v$, i.e. $F_0 R\,\nabla\cdot\mathbf{j}_{\rm vis} = \nabla\cdot(\mu R^3\nabla\omega) + \partial_R(2\mu R^2\omega)$, the physical form. The hyperviscosity $-\mu_{\rm num}\,\nabla^2 v\,\nabla^2\omega$ is a numerical current of the same family.

Key: the currents JOREK has

The vorticity equation (37) is $F_0 R\,\nabla\cdot\mathbf{J} = 0$ with

$$\mathbf{J}_\parallel = -\frac{j}{F_0}\,\mathbf{B} \qquad \text{(from the induction equation)}$$

$$\mathbf{j}_{\rm dia} = \frac{R}{F_0}\,\mathbf{e}_\varphi \times \nabla p = \frac{R}{F_0}(-\partial_Z p, \partial_R p) \qquad \text{(curvature term; } \nabla\cdot\mathbf{j}_{\rm dia} = -\tfrac{2}{F_0}\partial_Z p\text{)}$$

$$\mathbf{j}_{\rm pol} = -\frac{\rho R^2}{F_0}\left(\partial_t + \mathbf{v}_E\cdot\nabla + \dots\right)\nabla_\perp u \qquad \text{(lhs + Reynolds + gyroviscous)}$$

$$\mathbf{j}_{\rm vis} = \frac{\mu}{F_0}\,\nabla_\perp\omega \text{ (old setup, AUG case)}, \quad \simeq \frac{\mu R^2}{F_0}\,\nabla_\perp\omega \text{ (new)} \qquad \text{(viscosity, + hyperviscous)}$$

all in normalised units ($\mathbf{J}_{SI} = \mathbf{J}/\mu_0$). Missing relative to SOLPS: ion–neutral friction and the anomalous ohmic conductivity. Present in JOREK but not in SOLPS: the time-dependent polarisation current and the induction equation for $J_\parallel$.

Flux-surface average. Multiplying (37) by a flux function and integrating, term (b) vanishes identically ($[\psi,\, j]$ against $v'(\psi)\,[\psi,\, \psi]$), and term (c) vanishes when $p = p(\psi)$. This is the statement of Sec. 5: parallel and diamagnetic currents redistribute charge within a surface; only polarisation, viscosity and the pressure asymmetry carry net charge across it [13]. On open field lines, where p is not a flux function, term (c) does carry charge across, and that is the leg source of (15).

17 Boundary conditions for currents in JOREK

Plates (sheath row, `sheath_j`). The current variable j at the plate nodes is set by the characteristic (10) written in JOREK's variables:

$$j = j_{\rm sat}^{\rm JOREK}\, f(x), \qquad j_{\rm sat}^{\rm JOREK} \propto \frac{\rho\, c_s}{|B|} \quad \text{(so that } J_\parallel = j_{\rm sat} \text{ at the plate)}, \qquad x = \Lambda - \frac{a_n\,(u - C_V V_w)}{2\,T_e}, \tag{41}$$

with $a_n = 2eF_0\sqrt{\mu_0\rho_0}/m_i$ converting u to $e\Phi$ in units of the normalised temperature, and the ion flow at the plate fixed to the sound speed by the Mach-1 row on $V_\parallel$, so that $j_{\rm sat}$ is the same Bohm flux the flow carries. The value degree of freedom of u at those nodes is released: the vorticity equation then determines Φ and the row determines the current through it, exactly as BCPOT 11 does in SOLPS. The natural (surface) term of the u row carries the current j into the plate.

Flux-surface boundaries (types 2, 3: outer SOL edge, private-flux floor, corners). The u row at these nodes is replaced by the floating condition $\Phi = \Lambda T_e/e$ (`floating_u`) and j is held at its initial value. The replaced vorticity equation is not solved there, so whatever current (37) would need to send through that boundary is absorbed silently: its size is the residual of the dropped row. Summing the u value rows with the partition of unity gives the exact discrete identity

$$F_0\, I_{\rm plates}^{\rm par} = \sum_{\text{pinned/floating } u \text{ nodes}} r_u \;-\; S, \tag{42}$$

with r_u the residuals of the dropped rows and S the volume integral of the terms not in divergence form (the gyroviscous block). The plate current that does not return to the other plate is therefore a measured quantity only if r_u is printed; the postproc, which knows $J_\parallel$ and $\mathbf{j}_{\rm dia}$ but not $\mathbf{j}_{\rm pol}$ and $\mathbf{j}_{\rm vis}$, cannot see it.

Watch out: three different "currents" at the plate

`Jn_par` is the parallel current the sheath row sets and the only current the model transports into the wall. `Jnorm` is the pointwise parallel plus diamagnetic current from the fields, for information. The polarisation and viscous currents have no postproc expression at all. A budget made from postproc fields can therefore close only up to the part carried by $\mathbf{j}_{\mathrm{pol}} + \mathbf{j}_{\mathrm{vis}}$ through the boundaries.

18 Coefficients of the AUG #33616 case in SI

quantity	namelist	conversion	SI value
time unit	$n_0 = 0.7646 \times 10^{20}$, D	$t_0 = \sqrt{\mu_0 m_i n_0}$	5.67×10^{-7} s
resistivity on axis	`eta = 3e-8`	$\eta_{SI} = \mu_0 \eta_0 / t_0$	$6.7 \times 10^{-8}\,\Omega$m
resistivity at 30 eV (total)	`eta_T_dependent`, $T_0 = 7.3$ keV	$\times (T/T_0)^{-3/2}$	$2.5 \times 10^{-4}\,\Omega$m $\approx 20\times$ Spitzer
perpendicular viscosity	`visco = 2.3e-8`, `visco_T_dependent` (default on): $\mu \propto (T/T_0)^{-3/2}$, $T_0 = 7.3$ keV	$\nu_{SI} = \mathrm{visco}\,(T/T_0)^{-3/2}/(t_0\,\hat{\rho})$	$0.04/\hat{\rho}$ m^2/s on axis; $\times 150$ at 250 eV, $\times 3800$ at 30 eV: ~ 15 m^2/s in the separatrix layer, hundreds at the plates
hyper-resistivity	`eta_num = 3e-14`	numerical	–
hyper-viscosity	`visco_num = 2e-14`	numerical	–
sheath constant	`sheath_Lambda = 3`	–	3

Both coefficients follow the same $T^{-3/2}$ law anchored at the axis temperature, which is chosen for the pedestal dynamics. In the SOL this makes the parallel resistivity about twenty times Spitzer, which throttles the plate-to-plate current of Sec. 7 by that factor, and it makes the perpendicular viscosity one to three orders of magnitude larger than the anomalous value SOLPS uses ($D^{(AN)} \sim 0.3$–1 m^2/s), which feeds the viscous closure of Sec. 8. Both push the same way: away from the thermoelectric circuit and towards plate closing currents. Artola et al. (PPCF 63 (2021) 064004, Sec. 4.3) describe exactly this balance, $\mathbf{B} \cdot \nabla j \approx \mu_\perp \Delta\omega$, as the signature of a too large perpendicular viscosity under sheath current conditions, and prescribe lowering $\mu_\perp$ until the plate currents no longer depend on it.

19 SOLPS and JOREK side by side

	SOLPS (B2.5)	JOREK (model600)
charge conservation	finite-volume $\nabla\cdot\mathbf{J}=0$ for Φ	weak-form vorticity equation (37) for u
parallel current	algebraic Ohm's law, Spitzer/Zhdanov	induction equation (35), $\eta_0(T/T_0)^{-3/2}$, hyper-resistivity
diamagnetic / ∇B	effective $\tilde{j}^{(dia)}$, Eq. (28)	term (c), $[R^2, p]$, Eq. (38)
inertia	$j^{(in)}$ from $V_\parallel^2$; polarisation not time dependent	full $\rho\,\mathrm{d}_t\nabla_\perp\Phi$: time derivative, Reynolds stress, gyroviscous correction
viscosity	anomalous $\eta^{(AN)}$ on $V_\perp$, Eq. (30); parallel and heat viscosity currents	$\mu\nabla^2\omega$ with input `visco`; hyperviscosity
ion–neutral friction	yes	no
anomalous conductivity	$\sigma^{(AN)}$, optional	no
plate condition	sheath current as face flux, BCPOT 3/11	sheath row on j with released u value, Mach-1 on $V_\parallel$
flux-surface boundaries	prescribed potential/gradient/current, or physical $j_\perp$ let through (12, 13)	floating u row, j frozen; current absorbed as a row residual
global budget	exact, every boundary printed	exact by (42), boundary part not printed

Part V
Reading our results

20 The AUG #33616 axisymmetric run with the sheath row

At 16 ms of the $n=0$ phase with `sheath_j` and the thermoelectric Ohm's law (run B), the tools of these notes give:

quantity	inner	outer	both	comment
parallel current into the plate	+0.35 kA	+0.26 kA	+0.61 kA	ions at both
strike-point band (ψ_N 1.005–1.015)	−0.20 kA	−0.19 kA	−0.40 kA	electrons at both
wing band (ψ_N 1.02–1.04)	+0.11 kA	+0.42 kA	+0.53 kA	ions at both
through-current $(I_{\mathrm{out}}-I_{\mathrm{in}})/2$, wings			+0.15 kA	asymmetry of the loops
side current into the wing tube, by carrier			+0.59 kA	98 % non-diamagnetic, above the X-point
$\Phi_{\mathrm{out}}-\Phi_{\mathrm{in}}$ vs $\Lambda\Delta T_e/e$, wings			29 V vs 21 V	tube nearly open-circuit
parallel + diamagnetic current through types 2/3			0.00 kA	postproc current does not leave there

With Table 1 and Eq. (24) the reading is:

1. The plate currents are *plate closing currents*: same sign at both plates in every band, and in each tube the current arrives through the sides, not from the other plate.

2. They are of *type 2*: the side current enters above the X-point and is carried by the currents the postproc does not contain, polarisation and viscosity, while the diamagnetic current of the legs passes through the tubes without net deposit. The source region is the separatrix shear layer, all the way round poloidally. The driving mechanism is the single-plate loop of Sec. 8.

3. The *thermoelectric* current is at most the 0.15 kA asymmetry and probably less: the end-to-end potential difference equals $\Lambda\Delta T_e/e$ within a few volts, i.e. the circuit of Fig. 4 is open, consistent with a tube resistance of 20 times Spitzer (Sec. 18).

4. The net 0.61 kA leaves through the flux-surface boundaries as polarisation and viscous current, absorbed by the floating rows there, Eq. (42). Charge is conserved; the postproc cannot show it because it does not carry those currents.

Key: what to say

The plate currents of a self-consistent sheath simulation do not sum to zero and are not expected to. Each plate runs a radial circuit between its hot strike point and its cold wings, closed across the field in the SOL by the polarisation and viscous currents of the vorticity equation; both plates therefore carry the same sign. The plate-to-plate thermoelectric current is small because both plate ends float and the parallel path is resistive. The sum of the plate currents is the cross-field current through the outer SOL and private-flux boundaries, the analogue of SOLPS's radial boundary currents. Its magnitude is set by the perpendicular viscosity of the separatrix layer, which with the temperature-dependent viscosity of this case is far above the SOLPS anomalous value, and by the parallel resistivity through the competing thermoelectric loop; both are inputs, not predictions.

21 Checklist for a plate-current figure

1. Is the sign convention stated? (ions positive, both plates)

2. Do both plates obey the characteristic (10) pointwise? If not, the rest is meaningless.

3. Plot $j_n(\psi)$ of both plates on one axis. Same sign: closing currents. Opposite sign: through-currents.

4. Compare $\Phi_{\text{out}} - \Phi_{\text{in}}$ with $\Lambda\Delta T_e/e$ at matched ψ. Equal: open circuit. Larger: thermoelectric current flowing, estimate it with (23).

5. If the sum is not zero, say where it goes: side currents per tube (which carrier, where along the tube), and the boundary currents of the domain.

6. State the coefficients that set the result: $\eta_\parallel$ at the plates in units of Spitzer, $\nu_\perp$ in m^2/s.

A Reading list, annotated

1. **Stangeby**, *The Plasma Boundary of Magnetic Fusion Devices*, IoP 2000. Chapters 2 and 25: sheath, floating potential, the characteristic, and a first discussion of SOL currents.

2. **Staebler and Hinton**, Nucl. Fusion 29 (1989) 1820. The 1D thermoelectric circuit with sheath resistances; the origin of Eq. (23).

3. **Rozhansky et al.**, Nucl. Fusion 41 (2001) 387. Derivation of every perpendicular current in B2.5 and what sets the potential near and far from the separatrix. The PDF is in the repository root.

4. **Rozhansky et al.**, Nucl. Mater. Energy 25 (2020) 100840. Pfirsch–Schlüter, thermoelectric and plate closing currents, with SOLPS-ITER and probe comparisons.

5. **Schaffer, Chankin et al.**, Pfirsch–Schlüter currents in the JET divertor. The measurement that closing currents, not thermoelectric ones, can dominate a target current layer.

6. **Zhao et al.**, arXiv:2601.14749 (2026). The ∇B source of parallel current in the legs and the estimate that the thermoelectric current is ten times smaller in detached DIII-D.

7. **SOLPS-ITER manual**, Appendix C.6 (currents) and C.7.3 (potential boundary conditions); the file is in the repository root.

8. **Hölzl et al.**, Nucl. Fusion 61 (2021) 065001. The JOREK reduced-MHD equations and normalisation.

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[6] M. Zhao, T. Rognlien, B. Zhu, F. Scotti, X. Ma, A. McLean, "Triggers for plasma detachment bifurcation in the edge divertor region of tokamaks", arXiv:2601.14749 (2026).

[7] I. Senichenkov, E. Kaveeva, V. Rozhansky, E. Sytova, I. Veselova, S. Voskoboynikov, D. Coster, "Role of electric field in the formation of detached regime", EPS conference paper.

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[10] R. A. Pitts et al., Nucl. Fusion **43** (2003) 1145.

[11] D. P. Coster et al., IAEA Fusion Energy Conference paper EX/P4-20, SOLPS target currents versus ASDEX Upgrade.

[12] M. Hölzl et al., Nucl. Fusion **61** (2021) 065001.

[13] "What sets E_r in model600", repository note `doc/er_derivation/jorek_Er.pdf` (2026).

[14] "Ohm's law as assembled in model600", repository note `doc/elm_sheath_thermo/ohms_law.pdf` (2026).